BESTLINK COLLEGE OF THE PHILIPPINES
College of Computer Studies
Bachelor of Science in Information Technology

FUND HARMONY
Integrated Microfinance Management Platform

Project documentation Submitted to:

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In partial fullfillment of the requirements for
BPM101-BUSINESS PROCESS MANAGEMENT

Submitted by:

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March 2026

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CHAPTER 1 – INTRODUCTION

1.1 Background of the study

Microfinance institutions provide financial assistance to small businesses and low-income individuals who often have limited access to traditional banking services. These institutions help promote financial inclusion by offering loans, savings opportunities, and financial support to underserved communities.

However, many microfinance institutions still rely on manual record-keeping or disconnected systems for managing clients, loans, and payments. This can lead to delays in loan processing, inaccurate records, and difficulty in monitoring repayments and financial performance.

To address these challenges, **FundHarmony: Integrated Microfinance Management Platform** was developed. This system integrates key microfinance operations such as client management, loan application processing, repayment tracking, and financial reporting into a centralized web-based platform.

By automating these processes, the system helps institutions improve operational efficiency, reduce human errors, and provide real-time financial monitoring for better decision-making.

1.2 Objectives of the Study

1.2.1 General Objective

The main objective of this study is to develop a **web-based microfinance management system** that integrates client records, loan processing, repayment tracking, and reporting into a centralized platform.

1.2.2 Specific Objectives

1. To create a centralized database for storing client information.
2. To automate the loan application and approval process.
3. To track loan repayments and update balances automatically.
4. To generate real-time financial reports for monitoring operations.
5. To improve efficiency and accuracy in microfinance management.

1.3 Scope and Limitations

Scope

The FundHarmony system focuses on managing microfinance operations through a web-based platform. The system includes the following functions:

- Client registration and profile management
- Loan application processing
- Loan approval and repayment tracking
- Payment recording and loan balance updates
- Financial report generation for administrators

Limitations

The system has the following limitations:

- It is designed only for microfinance institutions.
- The system requires internet access to operate.
- Online payment gateway integration is not included in the current version.
- The system focuses only on loan management and does not include savings or investment services.

1.4 Significance of the Study

The development of the FundHarmony system provides benefits to several groups:

Microfinance Institutions
The system helps institutions manage loans, client data, and financial records efficiently.

Administrators and Staff
It simplifies administrative tasks such as client registration, loan approval, repayment monitoring, and report generation.

Clients / Borrowers
Clients can easily apply for loans, monitor their loan status, and track payment history through the system.

Researchers and Students
This study can serve as a reference for future research related to financial technology systems and microfinance management platforms.

Chapter 2 – Review of Related Literature / Systems

2.1 Review of Related Literature (RRL)

Microfinance systems are designed to support financial inclusion by providing financial services to individuals who lack access to traditional banking systems. According to Armendáriz and Morduch (2010), microfinance institutions help small entrepreneurs gain access to capital, which promotes economic growth and poverty reduction.

Digital systems in financial services have significantly improved operational efficiency. Hussain and Das (2018) explained that digitalization enables institutions to automate financial processes, improve accuracy, and provide better financial monitoring.

Similarly, Ledgerwood and White (2006) emphasized that integrated financial systems allow institutions to manage loans, payments, and client information effectively, which reduces operational errors and improves service delivery.

2.2 Review of Related Systems (RRS)

Several existing systems are related to the FundHarmony platform:

1. Mifos X
Mifos X is an open-source platform used by microfinance institutions for loan management, client tracking, and financial reporting. Similar to FundHarmony, it centralizes financial operations in one system.

2. Musoni Microfinance System
Musoni is a cloud-based microfinance platform designed to automate loan management, payment tracking, and client records.

3. Loan Management System (LMS)
Loan Management Systems are widely used by financial institutions to manage loan applications, approvals, and repayments digitally.

These systems inspired the design of FundHarmony by demonstrating the importance of centralized digital platforms in improving financial operations.

Chapter 3 – System Design and Methodology

3.1 System Architecture / Overview

FundHarmony follows a **web-based system architecture**, where users access the system through a browser while the application processes data through a centralized server and database.

The system consists of two main users:

- **Admin / Staff**
- **Clients**

The database stores client information, loan records, payment data, and reports.

3.2 Use Case Diagram

The Use Case Diagram illustrates how users interact with the system.

Main user actions include:

Admin

- Register client
- Approve loan
- Record repayment
- Generate reports

Client

- Login
- Apply for loan
- View loan status
- Make payments
- View payment history

3.3 Entity Relationship Diagram (ERD)

The ERD represents the database structure of the system.

Main entities include:

- **Clients**
- **Loans**
- **Payments**
- **Users**
- **Reports**

Each entity is connected through relationships that define how data is stored and accessed within the system.

3.4 Process Flow / Flowchart

The process flow shows the step-by-step operation of the system.

Example flow:

Client Login → Apply for Loan → Admin Reviews Application → Loan Approved → Client Makes Payment → System Updates Balance → Report Generated

3.5 Tools and Technologies Used

The system was developed using the following technologies:

Programming Language

- PHP
- JavaScript
- HTML / CSS

Database

- MySQL

Development Tools

- Visual Studio Code
- XAMPP Server
- phpMyAdmin

These tools were used to design, develop, and test the FundHarmony system.

Chapter 4 – System Implementation and Results

This chapter presents the system interface and main features of the FundHarmony platform.

Figure 4.1 Login Page

The login page allows users to enter their username and password to access the system securely.

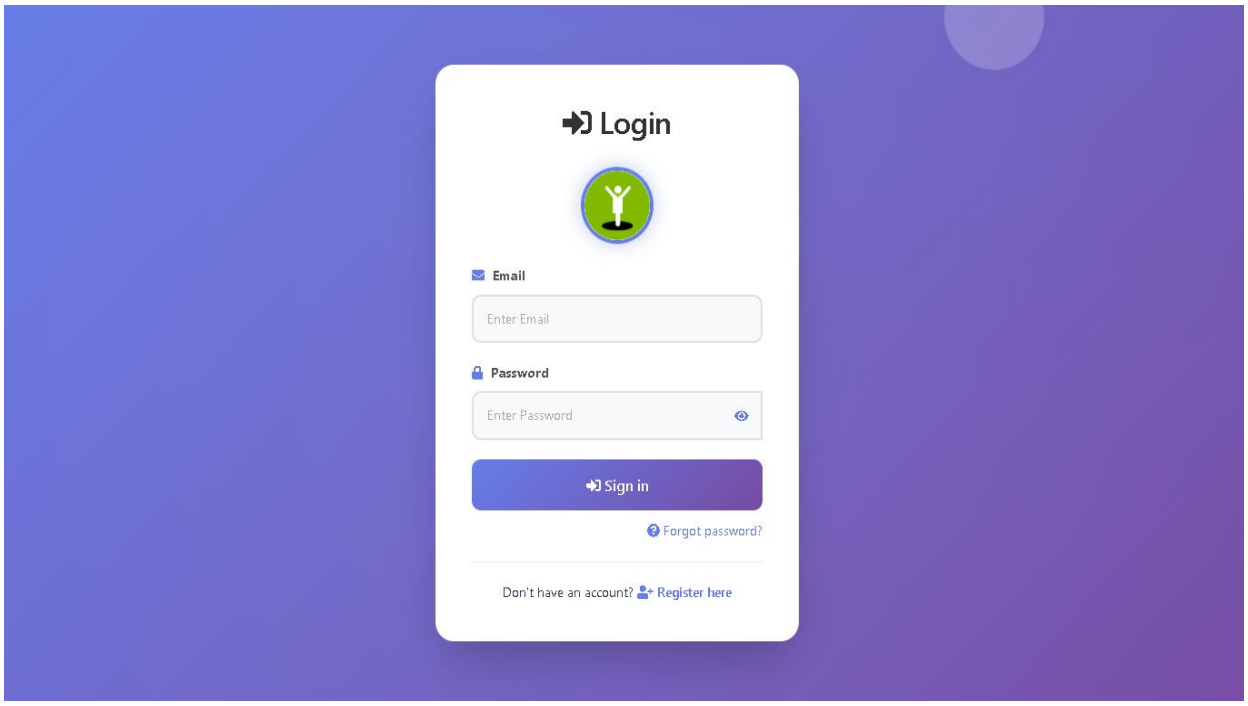


Figure 4.2 Dashboard

The dashboard displays system statistics such as total clients, active loans, and payment summaries.

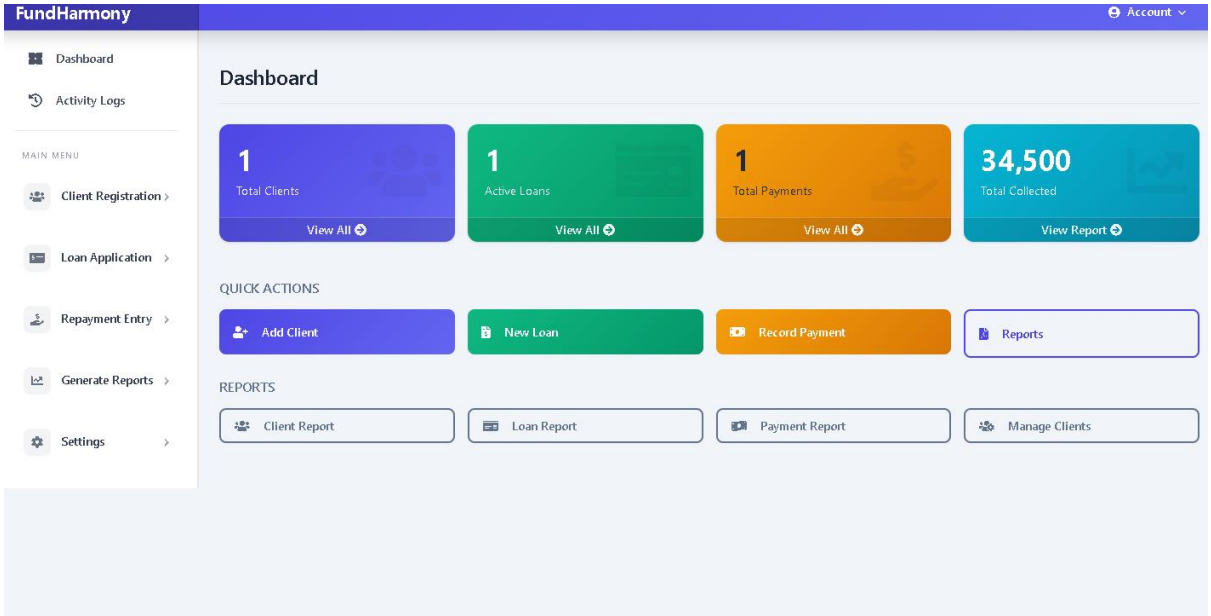
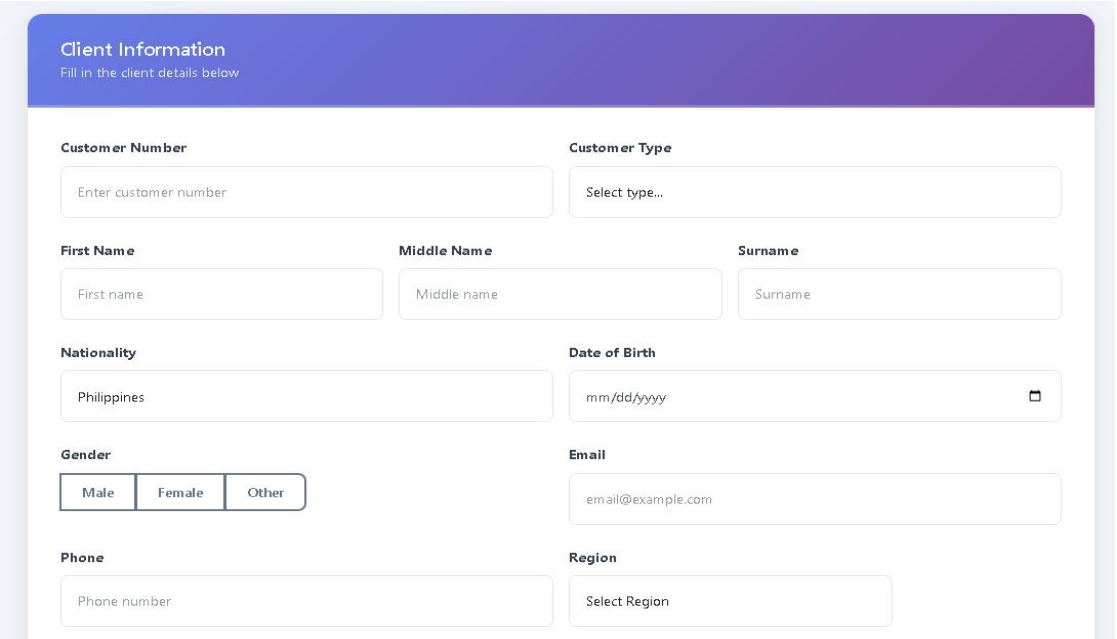


Figure 4.3 Client Registration Module

This module allows administrators to add new client records into the system database.



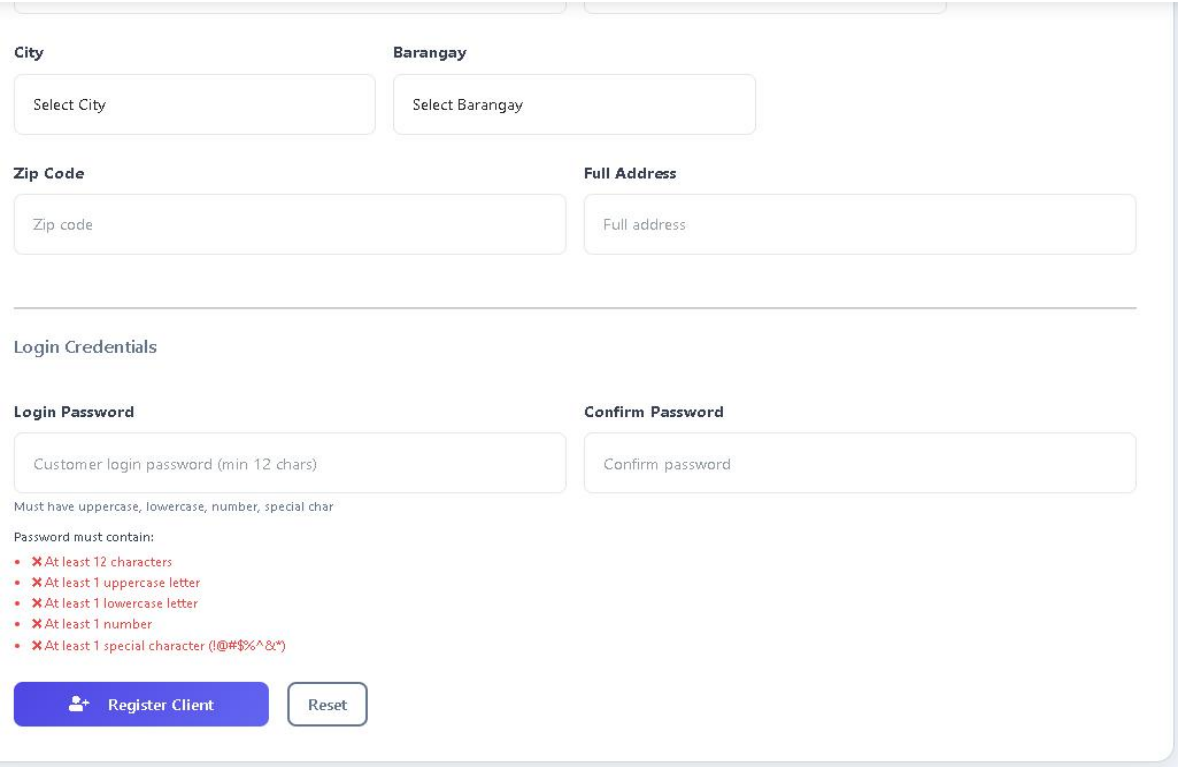


Figure 4.4 Loan Application Module

Clients can submit loan applications by entering loan amount, type, and repayment details.

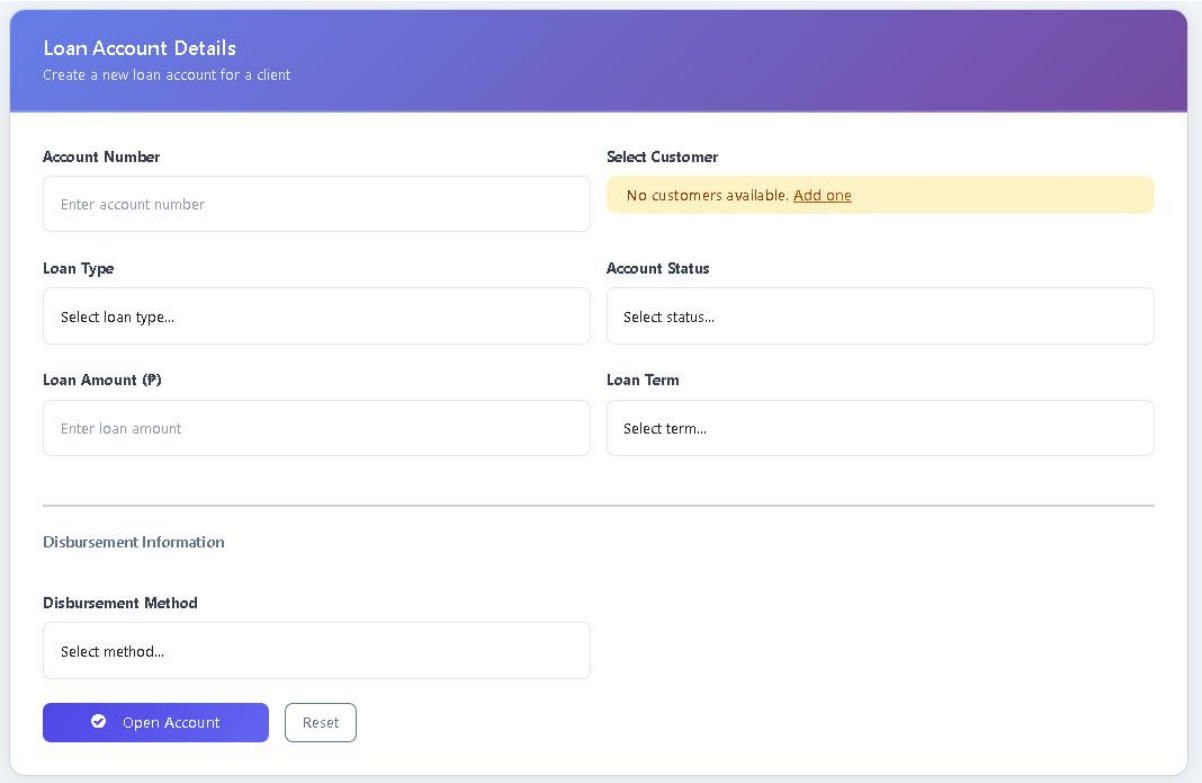


Figure 4.5 Repayment Tracking Module

This module records loan payments and automatically updates loan balances.

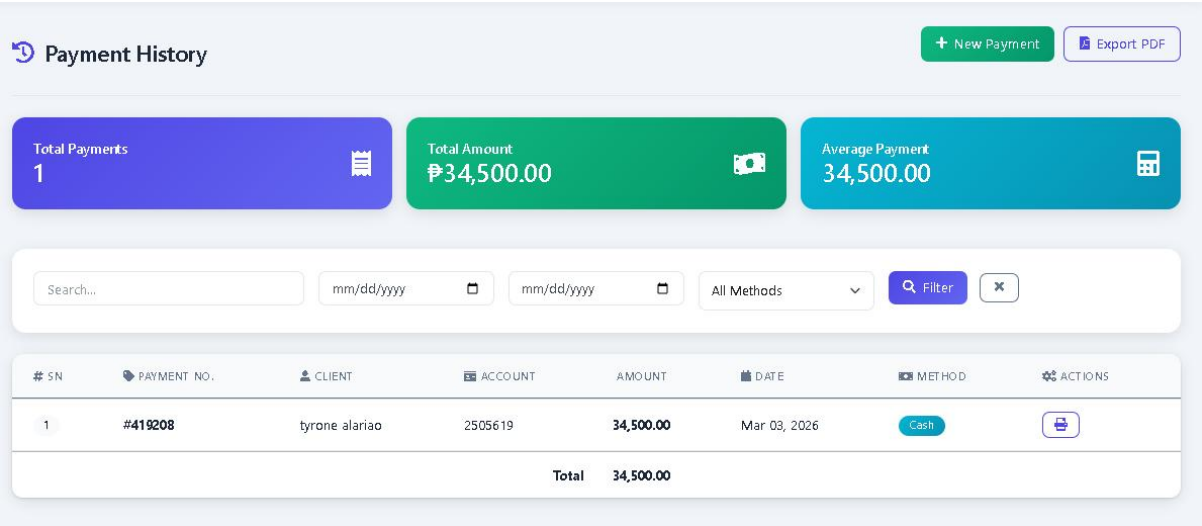


Figure 4.6 Report Generation

The system generates financial reports that help administrators monitor microfinance operations.

Payment Report

FundHarmony - Microfinance Management System

Generated on: 2026-03-08 07:06:59
All Payments

1	#419208	tyrone alariao	2505619	34,500.00	Mar 03, 2026	Cash

Loan Report

FundHarmony - Microfinance Management System

Generated on: 2026-03-08 07:06:30
All Loans

1	2505619	tyrone alariao	Emergency Loan	3 month(s)	?30,000.00	?4,500.00	?0.00	?0.00	Jun 02, 2026	Paid	Mar 02, 2026

Customer Report

All Customers

SN	Customer No.	Name	Email	Phone	Gender	DOB	Nationality	Date Added
1	859568034	tyrone alariao	tyronealariao06@gmail.com	09919094456	Male	May 10, 2003	Filipino	Mar 02, 2026

Chapter 5 – Conclusion and Recommendations

5.1 Conclusion

The FundHarmony system successfully provides a centralized platform for managing microfinance operations. By automating loan applications, client records, repayment tracking, and reporting, the system improves efficiency, accuracy, and financial monitoring. This helps microfinance institutions operate more effectively and provide better services to their clients.

5.2 Recommendations

For future improvements, the following enhancements are recommended:

Integration of online payment gateways

Mobile application support

Advanced data analytics and reporting tools

Improved security features such as two-factor authentication

These improvements can further enhance the performance and usability of the system.

REFERENCES

Armendáriz, B., & Morduch, J. (2010). *The Economics of Microfinance* (2nd ed.). MIT Press.
— A foundational book on how microfinance works and its impact on financial inclusion.

Ledgerwood, J., & White, V. (2006). *Transforming Microfinance Institutions: Providing Full Financial Services to the Poor*. World Bank.
— Discusses the importance of integrated systems and services in microfinance.

Schreiner, M. (2002). *Aspects of Outreach: A Framework for Discussion of the Social Benefits of Microfinance*. Journal of International Development, 14(5), 591–603.
— Provides insights into how microfinance services benefit clients socially and economically.

Hussain, M. A., & Das, N. (2018). Digitalization and the Future of Microfinance: Opportunities and Challenges. *International Journal of Economics and Financial Issues*, 8(4), 150–160.
— Explores how digital and web-based systems improve microfinance operations.

Mbogo, M. (2010). The Impact of Mobile Payments on the Success and Growth of Microbusiness: The Case of M-Pesa in Kenya. *Journal of Language, Technology & Entrepreneurship in Africa*, 2(1), 182–203.
— Shows an example of digital technology transforming financial services in the developing world.

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EDUCATION

Daraga Placer Masbate
Elementary School
2010-2015

Cataingan Masbate
Cataingan High
2016 - 2022

Bachelor of science in information technology
Bestlink College of the Philippines
2023 - 2025

SKILLS

- Management Skills
- Critical Thinking
- Communication Skills
- Negotiation
- Web development

LANGUAGE

- Filipino
- Cebuano
- English

PERSONAL INFO

- Age : 22
- Status: Single
- Gender : Male
- Religion: Catholic
- Date of birth: May 10, 2003
- Nationality: Filipino

PROFILE

Detail-oriented and self-motivated programmer with excellent problem-solving skills and a technical mindset. Collaborates effectively with others and has working knowledge of front-end programming languages, eager to support software engineering initiatives.

WORK EXPERIENCE

Quanta Corps — 2021 – 2022
Truck Helper/Assistant

- Assisted in loading and unloading goods safely and efficiently.
- Helped the driver with navigation and delivery coordination.

Sutukil Restaurant & Bar — 2022 - 2023
Crew Staff/Waiter

- Provided excellent customer service and assisted in taking orders.
- Prepared and served food and beverages following safety standards.
- Maintained cleanliness and organization of the dining and kitchen area.
- Handled cash transactions and operated the POS system when needed.
- Worked efficiently with team members to ensure customer satisfaction.

REFERENCES

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